BRACT's

Vishwakarma Institute of Information Technology, Pune Practical Implementation Sheet

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Subject Name: Deep Learning

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Problem Statement: Perform Image classification using Pretrained Vision Transformer model and compare its performance with CNN Model

Algorithm:

Step 1: Import the required libraries such as PyTorch, Torchvision, Transformers, NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn.

Step 2: Load the **CIFAR-10 image dataset** containing 10 different image classes such as airplane, automobile, bird, cat, dog, frog, horse, ship, and truck.

Step 3: Preprocess the CIFAR-10 images by converting them into tensors and normalizing the pixel values for CNN training.

Step 4: Split the dataset into **training, validation, and testing subsets** for model development and evaluation.

Step 5: Build a **CNN model** using convolutional layers, batch normalization, ReLU activation, pooling layers, dropout, and fully connected layers.

Step 6: Configure the CNN training parameters such as **learning rate, batch size, optimizer, loss function, and number of epochs**.

Step 7: Train the CNN model on the CIFAR-10 training dataset and evaluate its performance on the validation dataset.

Step 8: Load the pretrained **Vision Transformer (ViT)** model (google/vit-base-patch16-224-in21k) from Hugging Face and configure it for 10-class classification.

Step 9: Resize and preprocess the CIFAR-10 images to the input format required by the pretrained ViT, and fine-tune the ViT model on the training dataset.

Step 10: Evaluate both the **CNN and pretrained ViT models** on the test dataset using accuracy, precision, recall, and F1-score.

Step 11: Generate **confusion matrices and classification reports** for both models to analyze their class-wise performance.

Step 12: Compare the performance of the CNN and ViT models using a comparison table and graphical representation of their evaluation metrics.

Step 13: Display sample test images along with their **actual labels, CNN predictions, and ViT predictions** to visually compare the classification results.

Step 14: Determine the better-performing model based on the obtained **test accuracy and F1-score**, and analyze the advantages and limitations of CNN and ViT.

Step 15: Save the trained models and relevant results for **future use and deployment**.

Code:

```
# Imports
import os
import time
import random
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, random_split
from torchvision import datasets, transforms

from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    classification_report, confusion_matrix
)

from transformers import (
    ViTImageProcessor,
    ViTForImageClassification,
    TrainingArguments,
    Trainer
)

from PIL import Image

# Reproducibility
SEED = 42
```

```
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
if torch.cuda.is_available():
    torch.cuda.manual_seed_all(SEED)

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Device:", device)
print("PyTorch version:", torch.__version__)

"""## 1. Load the CIFAR-10 Dataset"""

# CIFAR-10 class names
class_names = [
    "airplane", "automobile", "bird", "cat", "deer",
    "dog", "frog", "horse", "ship", "truck"
]

# Basic transform for CNN
cnn_transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize(
        mean=(0.4914, 0.4822, 0.4465),
        std=(0.2470, 0.2435, 0.2616)
    )
])

train_full = datasets.CIFAR10(
    root="./data",
    train=True,
    download=True,
    transform=cnn_transform
)

test_dataset = datasets.CIFAR10(
    root="./data",
    train=False,
    download=True,
    transform=cnn_transform
)

# Use 45,000 training and 5,000 validation images
train_dataset, val_dataset = random_split(
    train_full,
    [45000, 5000],
    generator=torch.Generator().manual_seed(SEED)
```

```
)

BATCH_SIZE = 128

train_loader = DataLoader(
    train_dataset, batch_size=BATCH_SIZE, shuffle=True,
    num_workers=2, pin_memory=torch.cuda.is_available()
)

val_loader = DataLoader(
    val_dataset, batch_size=BATCH_SIZE, shuffle=False,
    num_workers=2, pin_memory=torch.cuda.is_available()
)

test_loader = DataLoader(
    test_dataset, batch_size=BATCH_SIZE, shuffle=False,
    num_workers=2, pin_memory=torch.cuda.is_available()
)

print("Training images:", len(train_dataset))
print("Validation images:", len(val_dataset))
print("Test images:", len(test_dataset))

# Display sample images
images, labels = next(iter(train_loader))

plt.figure(figsize=(12, 8))
for i in range(12):
    img = images[i].permute(1, 2, 0).numpy()
    img = img * np.array((0.2470, 0.2435, 0.2616)) + np.array((0.4914, 0.4822, 0.4465))
    img = np.clip(img, 0, 1)

    plt.subplot(3, 4, i + 1)
    plt.imshow(img)
    plt.title(class_names[labels[i].item()])
    plt.axis("off")

plt.tight_layout()
plt.show()

"""## 2. Build and Train the CNN Model"""

class CNNModel(nn.Module):
    def __init__(self, num_classes=10):
        super().__init__()
```

```
        self.features = nn.Sequential(
            nn.Conv2d(3, 32, kernel_size=3, padding=1),
            nn.BatchNorm2d(32),
            nn.ReLU(),
            nn.MaxPool2d(2),

            nn.Conv2d(32, 64, kernel_size=3, padding=1),
            nn.BatchNorm2d(64),
            nn.ReLU(),
            nn.MaxPool2d(2),

            nn.Conv2d(64, 128, kernel_size=3, padding=1),
            nn.BatchNorm2d(128),
            nn.ReLU(),
            nn.MaxPool2d(2),

            nn.Conv2d(128, 256, kernel_size=3, padding=1),
            nn.BatchNorm2d(256),
            nn.ReLU(),
            nn.AdaptiveAvgPool2d((1, 1))
        )

        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Dropout(0.4),
            nn.Linear(256, 128),
            nn.ReLU(),
            nn.Dropout(0.3),
            nn.Linear(128, num_classes)
        )

    def forward(self, x):
        x = self.features(x)
        return self.classifier(x)


cnn_model = CNNModel().to(device)
print(cnn_model)

criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(cnn_model.parameters(), lr=0.001)
scheduler = optim.lr_scheduler.ReduceLROnPlateau(
    optimizer, mode="min", factor=0.5, patience=2
)

def train_cnn(model, train_loader, val_loader, epochs=5):
```

```
history = {
    "train_loss": [],
    "train_acc": [],
    "val_loss": [],
    "val_acc": []
}

best_val_acc = 0.0
best_state = None
start_time = time.time()

for epoch in range(epochs):
    model.train()
    running_loss = 0.0
    correct = 0
    total = 0

    for inputs, targets in train_loader:
        inputs, targets = inputs.to(device), targets.to(device)

        optimizer.zero_grad()
        outputs = model(inputs)
        loss = criterion(outputs, targets)
        loss.backward()
        optimizer.step()

        running_loss += loss.item() * inputs.size(0)
        correct += (outputs.argmax(1) == targets).sum().item()
        total += targets.size(0)

    train_loss = running_loss / total
    train_acc = correct / total

    # Validation
    model.eval()
    val_loss_sum = 0.0
    val_correct = 0
    val_total = 0

    with torch.no_grad():
        for inputs, targets in val_loader:
            inputs, targets = inputs.to(device), targets.to(device)
            outputs = model(inputs)
            loss = criterion(outputs, targets)

            val_loss_sum += loss.item() * inputs.size(0)
```

```
            val_correct += (outputs.argmax(1) == targets).sum().item()
            val_total += targets.size(0)

        val_loss = val_loss_sum / val_total
        val_acc = val_correct / val_total

        scheduler.step(val_loss)

        history["train_loss"].append(train_loss)
        history["train_acc"].append(train_acc)
        history["val_loss"].append(val_loss)
        history["val_acc"].append(val_acc)

        if val_acc > best_val_acc:
            best_val_acc = val_acc
            best_state = {k: v.cpu().clone() for k, v in model.state_dict().items()}

        print(
            f"Epoch {epoch+1}/{epochs} | "
            f"Train Loss: {train_loss:.4f} | Train Acc: {train_acc:.4f} | "
            f"Val Loss: {val_loss:.4f} | Val Acc: {val_acc:.4f}"
        )

    elapsed = time.time() - start_time

    if best_state is not None:
        model.load_state_dict(best_state)

    return history, elapsed


CNN_EPOCHS = 5
cnn_history, cnn_train_time = train_cnn(
    cnn_model, train_loader, val_loader, epochs=CNN_EPOCHS
)

print(f"\nCNN training time: {cnn_train_time/60:.2f} minutes")

# Plot CNN training history
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.plot(cnn_history["train_loss"], label="Training Loss")
plt.plot(cnn_history["val_loss"], label="Validation Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
```

```
plt.title("CNN Loss")
plt.legend()

plt.subplot(1, 2, 2)
plt.plot(cnn_history["train_acc"], label="Training Accuracy")
plt.plot(cnn_history["val_acc"], label="Validation Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.title("CNN Accuracy")
plt.legend()

plt.tight_layout()
plt.show()

def evaluate_torch_model(model, loader):
    model.eval()
    all_preds = []
    all_labels = []

    with torch.no_grad():
        for inputs, labels in loader:
            inputs = inputs.to(device)
            outputs = model(inputs)
            preds = outputs.argmax(dim=1).cpu().numpy()

            all_preds.extend(preds)
            all_labels.extend(labels.numpy())

    return np.array(all_labels), np.array(all_preds)


cnn_y_true, cnn_y_pred = evaluate_torch_model(cnn_model, test_loader)

cnn_accuracy = accuracy_score(cnn_y_true, cnn_y_pred)
cnn_precision = precision_score(cnn_y_true, cnn_y_pred, average="weighted", zero_division=0)
cnn_recall = recall_score(cnn_y_true, cnn_y_pred, average="weighted", zero_division=0)
cnn_f1 = f1_score(cnn_y_true, cnn_y_pred, average="weighted", zero_division=0)

print("CNN Performance")
print(f"Accuracy : {cnn_accuracy:.4f}")
print(f"Precision: {cnn_precision:.4f}")
print(f"Recall   : {cnn_recall:.4f}")
print(f"F1-score : {cnn_f1:.4f}")

print("\nClassification Report:")
print(classification_report(
```

```
    cnn_y_true, cnn_y_pred,
    target_names=class_names,
    digits=4
))

# CNN confusion matrix
cm_cnn = confusion_matrix(cnn_y_true, cnn_y_pred)

plt.figure(figsize=(10, 8))
sns.heatmap(
    cm_cnn, annot=True, fmt="d", cmap="Blues",
    xticklabels=class_names, yticklabels=class_names
)
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.title("CNN Confusion Matrix")
plt.show()

"""## 3. Prepare CIFAR-10 for the Pretrained Vision Transformer

We use **ViT-Base-Patch16-224** pretrained on ImageNet-21k. CIFAR-10 images are resized to 224×224 and normalized using the processor expected by the pretrained model.
"""

# Hugging Face Vision Transformer setup
MODEL_NAME = "google/vit-base-patch16-224-in21k"

processor = ViTImageProcessor.from_pretrained(MODEL_NAME)

# Create a separate raw CIFAR-10 dataset for ViT.
# No normalization here; the processor performs the required preprocessing.
vit_train_full = datasets.CIFAR10(
    root="./data",
    train=True,
    download=False,
    transform=None
)

vit_test_dataset = datasets.CIFAR10(
    root="./data",
    train=False,
    download=False,
    transform=None
)

vit_train_dataset, vit_val_dataset = random_split(
```

```
    vit_train_full,
    [45000, 5000],
    generator=torch.Generator().manual_seed(SEED)
)

print("ViT training images:", len(vit_train_dataset))
print("ViT validation images:", len(vit_val_dataset))
print("ViT test images:", len(vit_test_dataset))

# Convert CIFAR-10 samples into the format expected by Hugging Face Trainer
class CIFAR10ViTDataset(torch.utils.data.Dataset):
    def __init__(self, subset, processor):
        self.subset = subset
        self.processor = processor

    def __len__(self):
        return len(self.subset)

    def __getitem__(self, idx):
        image, label = self.subset[idx]

        # CIFAR-10 returns a PIL image when transform=None
        encoded = self.processor(images=image, return_tensors="pt")

        return {
            "pixel_values": encoded["pixel_values"].squeeze(0),
            "labels": label
        }


vit_train = CIFAR10ViTDataset(vit_train_dataset, processor)
vit_val = CIFAR10ViTDataset(vit_val_dataset, processor)
vit_test = CIFAR10ViTDataset(vit_test_dataset, processor)

sample = vit_train[0]
print("Pixel values shape:", sample["pixel_values"].shape)
print("Label:", sample["labels"], class_names[sample["labels"]])

# Load pretrained ViT and replace the classifier for CIFAR-10
id2label = {i: name for i, name in enumerate(class_names)}
label2id = {name: i for i, name in enumerate(class_names)}

vit_model = ViTForImageClassification.from_pretrained(
    MODEL_NAME,
    num_labels=10,
    id2label=id2label,
```

```
    label2id=label2id,
    ignore_mismatched_sizes=True
)

vit_model = vit_model.to(device)

print("ViT model loaded.")
print("Number of parameters:",
      sum(p.numel() for p in vit_model.parameters()))

# Metrics for Hugging Face Trainer
def compute_metrics(eval_pred):
    predictions, labels = eval_pred
    predictions = np.argmax(predictions, axis=1)

    return {
        "accuracy": accuracy_score(labels, predictions),
        "precision": precision_score(
            labels, predictions, average="weighted", zero_division=0
        ),
        "recall": recall_score(
            labels, predictions, average="weighted", zero_division=0
        ),
        "f1": f1_score(
            labels, predictions, average="weighted", zero_division=0
        )
    }


# Training configuration
# For CPU, reduce epochs/batch size if training is too slow.
vit_batch_size = 16 if torch.cuda.is_available() else 4

training_args = TrainingArguments(
    output_dir="./vit_cifar10_results",
    eval_strategy="epoch",
    save_strategy="epoch",
    logging_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=vit_batch_size,
    per_device_eval_batch_size=vit_batch_size,
    num_train_epochs=2,
    weight_decay=0.01,
    load_best_model_at_end=True,
    metric_for_best_model="accuracy",
    greater_is_better=True,
```

```
    remove_unused_columns=False,
    report_to="none",
    fp16=torch.cuda.is_available()
)

trainer = Trainer(
    model=vit_model,
    args=training_args,
    train_dataset=vit_train,
    eval_dataset=vit_val,
    compute_metrics=compute_metrics
)

print("Trainer is ready.")

# Fine-tune pretrained ViT
print("Starting ViT fine-tuning...")
vit_start = time.time()

train_result = trainer.train()

vit_train_time = time.time() - vit_start

print(f"\nViT training time: {vit_train_time/60:.2f} minutes")

# Evaluate ViT on the test set
vit_test_output = trainer.predict(vit_test)

vit_y_true = vit_test_output.label_ids
vit_y_pred = np.argmax(vit_test_output.predictions, axis=1)

vit_accuracy = accuracy_score(vit_y_true, vit_y_pred)
vit_precision = precision_score(vit_y_true, vit_y_pred, average="weighted", zero_division=0)
vit_recall = recall_score(vit_y_true, vit_y_pred, average="weighted", zero_division=0)
vit_f1 = f1_score(vit_y_true, vit_y_pred, average="weighted", zero_division=0)

print("Vision Transformer Performance")
print(f"Accuracy : {vit_accuracy:.4f}")
print(f"Precision: {vit_precision:.4f}")
print(f"Recall   : {vit_recall:.4f}")
print(f"F1-score : {vit_f1:.4f}")

print("\nClassification Report:")
print(classification_report(
    vit_y_true, vit_y_pred,
    target_names=class_names,
```

```
    digits=4
))

# ViT confusion matrix
cm_vit = confusion_matrix(vit_y_true, vit_y_pred)

plt.figure(figsize=(10, 8))
sns.heatmap(
    cm_vit, annot=True, fmt="d", cmap="Greens",
    xticklabels=class_names, yticklabels=class_names
)
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.title("Vision Transformer Confusion Matrix")
plt.show()

"""## 4. Compare CNN and Vision Transformer Performance"""

comparison = pd.DataFrame({
    "Model": ["CNN", "Pretrained ViT"],
    "Accuracy": [cnn_accuracy, vit_accuracy],
    "Precision": [cnn_precision, vit_precision],
    "Recall": [cnn_recall, vit_recall],
    "F1-Score": [cnn_f1, vit_f1],
    "Training Time (min)": [
        cnn_train_time / 60,
        vit_train_time / 60
    ]
})

comparison.style.format({
    "Accuracy": "{:.4f}",
    "Precision": "{:.4f}",
    "Recall": "{:.4f}",
    "F1-Score": "{:.4f}",
    "Training Time (min)": "{:.2f}"
})

# Visual comparison
metrics = ["Accuracy", "Precision", "Recall", "F1-Score"]

x = np.arange(len(metrics))
width = 0.35

plt.figure(figsize=(10, 6))
plt.bar(x - width/2, comparison.loc[0, metrics].values.astype(float),
```

```
        width, label="CNN")
plt.bar(x + width/2, comparison.loc[1, metrics].values.astype(float),
        width, label="Pretrained ViT")

plt.xticks(x, metrics)
plt.ylabel("Score")
plt.ylim(0, 1.05)
plt.title("CNN vs Pretrained Vision Transformer")
plt.legend()
plt.grid(axis="y", alpha=0.3)
plt.show()

# Determine the better model
best_model = comparison.loc[comparison["Accuracy"].idxmax(), "Model"]
best_accuracy = comparison["Accuracy"].max()

print("Best model based on test accuracy:", best_model)
print(f"Best test accuracy: {best_accuracy:.4f}")

print("\nAccuracy improvement of ViT over CNN:",
      f"{(vit_accuracy - cnn_accuracy) * 100:.2f} percentage points")

"""## 5. Test Predictions — Visual Comparison"""

# Display selected test images with CNN and ViT predictions

# Get raw images directly from CIFAR-10
raw_test = datasets.CIFAR10(
    root="./data",
    train=False,
    download=False,
    transform=None
)

indices = [0, 1, 2, 3, 4, 5, 6, 7]

plt.figure(figsize=(14, 8))

for plot_idx, idx in enumerate(indices):
    image, true_label = raw_test[idx]

    # CNN prediction
    cnn_input = cnn_transform(image).unsqueeze(0).to(device)
    with torch.no_grad():
        cnn_output = cnn_model(cnn_input)
        cnn_pred = cnn_output.argmax(1).item()
```

```
    # ViT prediction
    vit_input = processor(images=image, return_tensors="pt")
    vit_input = {k: v.to(device) for k, v in vit_input.items()}

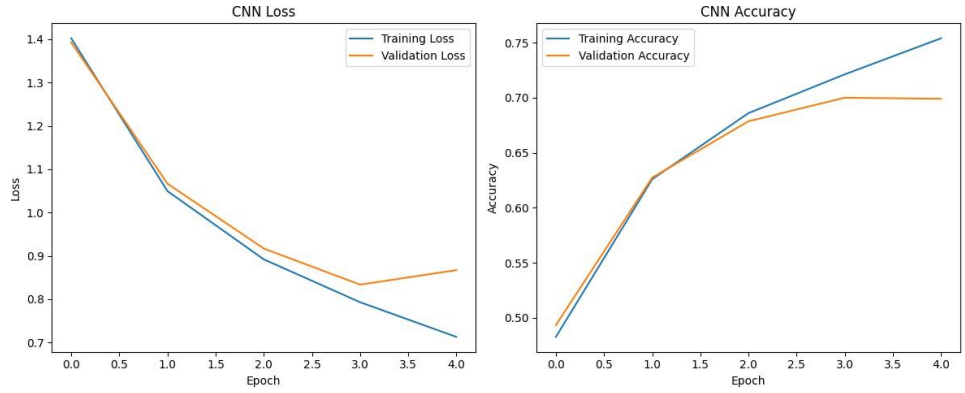
    with torch.no_grad():
        vit_output = vit_model(**vit_input)
        vit_pred = vit_output.logits.argmax(1).item()

    plt.subplot(2, 4, plot_idx + 1)
    plt.imshow(image)
    plt.title(
        f"True: {class_names[true_label]}\n"
        f"CNN: {class_names[cnn_pred]}\n"
        f"ViT: {class_names[vit_pred]}",
        fontsize=9
    )
    plt.axis("off")

plt.tight_layout()
plt.show()
```

Output:

```
Epoch 1/5 | Train Loss: 1.4022 | Train Acc: 0.4828 | Val Loss: 1.3926 | Val Acc: 0.4934
Epoch 2/5 | Train Loss: 1.0496 | Train Acc: 0.6260 | Val Loss: 1.0666 | Val Acc: 0.6274
Epoch 3/5 | Train Loss: 0.8921 | Train Acc: 0.6860 | Val Loss: 0.9171 | Val Acc: 0.6786
Epoch 4/5 | Train Loss: 0.7932 | Train Acc: 0.7212 | Val Loss: 0.8337 | Val Acc: 0.7000
Epoch 5/5 | Train Loss: 0.7131 | Train Acc: 0.7540 | Val Loss: 0.8672 | Val Acc: 0.6990
```

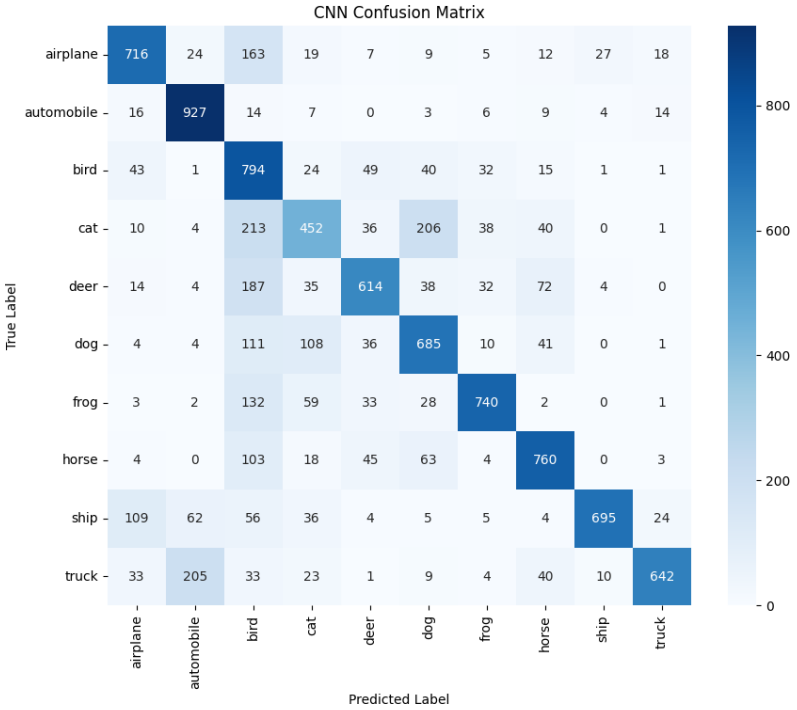


```
CNN Performance
Accuracy : 0.7025
Precision: 0.7354
Recall   : 0.7025
F1-score : 0.7069

Classification Report:
              precision    recall  f1-score   support

    airplane     0.7521    0.7160    0.7336      1000
  automobile     0.7518    0.9270    0.8303      1000
        bird     0.4396    0.7940    0.5659      1000
         cat     0.5787    0.4520    0.5076      1000
        deer     0.7442    0.6140    0.6729      1000
         dog     0.6308    0.6850    0.6568      1000
        frog     0.8447    0.7400    0.7889      1000
       horse     0.7638    0.7600    0.7619      1000
        ship     0.9379    0.6950    0.7984      1000
       truck     0.9106    0.6420    0.7531      1000

    accuracy                         0.7025     10000
   macro avg     0.7354    0.7025    0.7069     10000
weighted avg     0.7354    0.7025    0.7069     10000
```



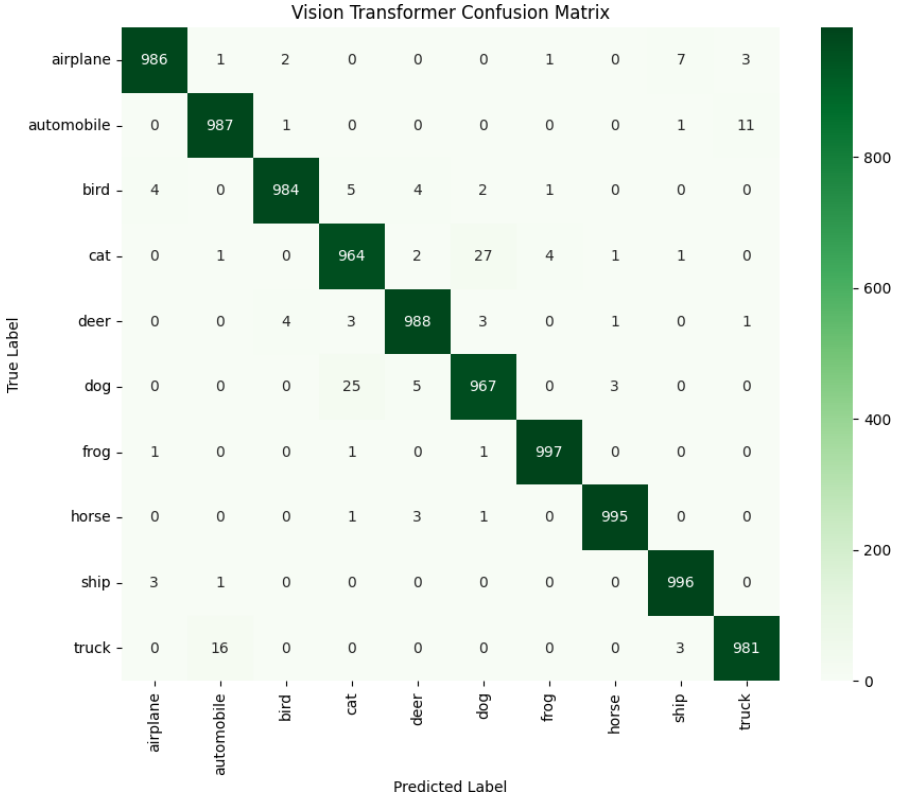
Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	0.295406	0.087928	0.981200	0.981504	0.981200	0.981184
2	0.032078	0.064822	0.983400	0.983458	0.983400	0.983403

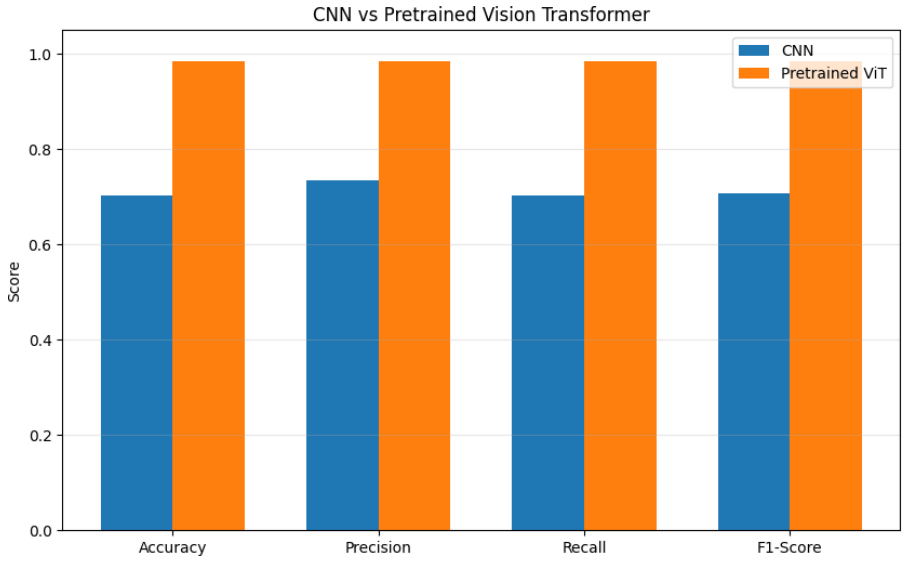
```
Vision Transformer Performance
Accuracy : 0.9845
Precision: 0.9845
Recall   : 0.9845
F1-score : 0.9845

Classification Report:
              precision    recall  f1-score   support

    airplane     0.9920    0.9860    0.9890      1000
  automobile     0.9811    0.9870    0.9840      1000
        bird     0.9929    0.9840    0.9884      1000
         cat     0.9650    0.9640    0.9645      1000
        deer     0.9860    0.9880    0.9870      1000
         dog     0.9660    0.9670    0.9665      1000
        frog     0.9940    0.9970    0.9955      1000
       horse     0.9950    0.9950    0.9950      1000
        ship     0.9881    0.9960    0.9920      1000
       truck     0.9849    0.9810    0.9830      1000

    accuracy                         0.9845     10000
   macro avg     0.9845    0.9845    0.9845     10000
weighted avg     0.9845    0.9845    0.9845     10000
```




CNN vs Pretrained Vision Transformer
CNN
Pretrained ViT
Score
1.0
0.8
0.6
0.4
0.2
0.0
Accuracy
Precision
Recall
F1-Score


True: cat
CNN: cat
ViT: cat
True: ship
CNN: automobile
ViT: ship
True: ship
CNN: automobile
ViT: ship
True: airplane
CNN: airplane
ViT: airplane
True: frog
CNN: frog
ViT: frog
True: frog
CNN: frog
ViT: frog
True: automobile
CNN: automobile
ViT: automobile
True: frog
CNN: bird
ViT: frog